

The AI Clone as Extended Self: How an AI Clone's Emergent Prosocial Behavior Shapes Its Owner's Self-Perception

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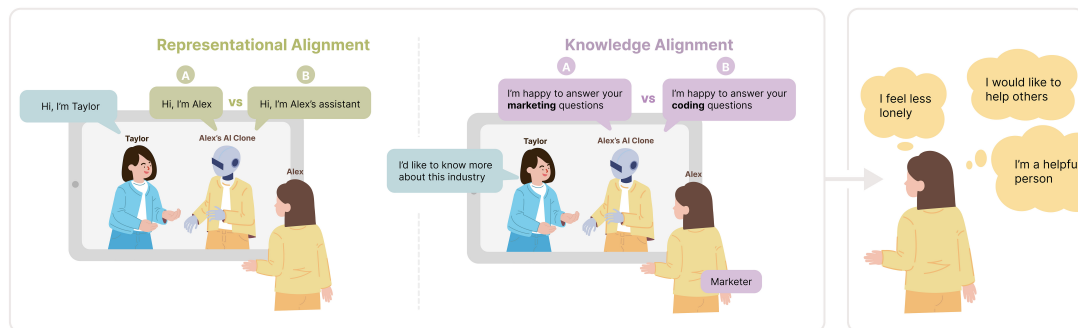


Fig. 1. **A visual overview of our study.** We investigated how observing an AI clone's prosocial behavior can influence its owner's self-perception of helpfulness, well-being, and prosocial willingness. Furthermore, we explored how **representational alignment** and **knowledge alignment** between the AI clone and its owner further shape these outcomes.

As AI-powered digital representations that capture certain aspects of a person, known as AI clones, become more prevalent, a critical question emerges: how do AI clones' emergent social behaviors influence their owners' self-perception? Such influences may present both benefits like enhanced self-image and risks such as eroding personal identity. To unpack this influence, we conducted a controlled experiment ($N = 159$) where participants observed their AI clone engaging in prosocial behavior. Results showed that merely watching their clone help another person increased participants' self-esteem and prosocial willingness while reducing feelings of loneliness. Moreover, clones that spoke in the first person and provided informational help aligned with participants' knowledge more effectively reduced loneliness and boosted prosocial willingness. Surprisingly, when clones offered help beyond participants' knowledge, participants reported greater self-perceptions of helpfulness. We discuss implications for designing AI clones that can benefit users while reducing potential harms.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**.

Additional Key Words and Phrases: Prosocial Behaviors, Helping, Well-Being, Human Basic Psychological Needs, Human-AI Interaction

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1 Introduction

Recently, a Chinese live streamer created an AI clone of himself to host a shopping stream. The event attracted 13 million viewers and generated over ¥55 million in sales [18]. AI clones, once confined to science fiction, are now a reality. Broadly, AI clones are digital entities that use AI to represent some aspects of a real individual [39]. People can have AI clones represent them in meetings [40] or handle tasks when they are multitasking [19]. Yet, as AI clones gain popularity and AI agents increasingly demonstrate emergent social behaviors [51], a critical question emerges: how will a person be influenced by their AI clone’s emergent social behavior, particularly their self-perception? Answering this is crucial, as the answer may present a duality of risk and opportunity. On one hand, such influence may lead to identity crises, as behaviors by an AI clone that diverge from the owner may undermine the owner’s self-concept [39]; on the other, it might be harnessed to enhance the owner’s self-perception, which is known to foster more positive actions and outcomes [15, 30, 43]. Despite these possibilities, this influence has not yet been empirically tested.

The theory of vicarious self-perception offers a lens through which to understand this potential influence. The theory posits that people sometimes internalize the attributes of others with whom they feel a “sense of overlap” [30]. For example, when participants were led to believe their brainwaves overlapped with a stranger’s, they rated themselves as more helpful and acted more prosocially after observing that stranger perform a helpful act [30]. This provides a theoretical basis for our inquiry and points to a particular social behavior worth attention: prosocial behavior. Prosocial behavior is fundamental to human social life and survival [57], and since AI social behaviors are often modeled on human interactions [24], it is plausible that AI clones can act prosocially. Furthermore, prosocial behavior is known to enhance the well-being of the help provider [63], a dimension overlooked in prior vicarious self-perception research [30]. Therefore, by focusing on an AI clone’s prosocial behavior, it becomes possible to test not only if vicarious self-perception occurs in this new context, but also the novel question of whether a clone’s actions can vicariously enhance its owner’s well-being.

When exploring this question, notably, a further factor to consider is the individual’s “sense of overlap” with the other agent, as this is what allows for the internalization of observed behaviors [30]. In the context of AI clones, this sense of overlap may be shaped by two design dimensions: representation and knowledge. “Representation” refers to whether the AI clone uses a first-person or third-person perspective [40, 47]. While this is an ongoing debate, research has found that first-person clones are often perceived as better representatives of the owner [40]. It is therefore likely that clone owners may also feel a stronger sense of overlap with first-person clones. In terms of knowledge, this refers to whether the AI clone’s knowledge aligns with the owner’s. While more knowledge is typically better in AI design [6], a clone that knows more than its owner could be unsettling [39]. Therefore, in the context of AI clones, it is likely that greater knowledge alignment can foster a stronger sense of overlap between individuals and their clones. However, these remain hypotheses in need of empirical testing.

To examine these questions, we conducted a randomized online experiment ($N = 159$). We first collected participants’ information to create their AI clones. Later, we invited participants to return to observe a conversation between their clone and what they believed was another participant, but was in fact a scripted character. Specifically, participants were randomly assigned to conditions where their clone either engaged in simple small talk (control) or also performed a prosocial act by providing informational help. For the prosocial conditions, we further manipulated the clone’s representational alignment (first-person vs. third-person) and its knowledge alignment (aligned vs. misaligned with the participant). Our results showed that participants who observed their clone act prosocially experienced greater reductions in loneliness and increases in self-esteem and prosocial willingness. Clones that spoke in the first person

and whose informational help aligned with participants’ knowledge more effectively reduced participants’ loneliness and increased their prosocial willingness. Surprisingly, participants’ self-perception of helpfulness improved the most when their clones provided help that went beyond their knowledge.

This research makes several contributions. First, it provides the empirical evidence that an AI clone’s emergent social behavior can vicariously influence its clone owner, moving the discussion from speculation to empirical grounding. Second, we introduce the concept of the “AI clone as an extended self” and uncover its dual potential: AI clones can act as enhancers that improve a person’s well-being and self-perception, or as distorters that harm their self-image and create unrealistic social expectations. Finally, our findings yield concrete design guidelines that can inform future development. These guidelines demonstrate how to design AI clones for different purposes, ensuring this technology can be leveraged to better serve human needs.

2 Related Work

2.1 AI Clones

AI clones, digital entities that represent some aspects of real individuals through AI [39], are rapidly emerging. While AI clones are widely known for malicious uses like deepfakes, which use visual manipulations to create realistic but false depictions of people [64], researchers are increasingly exploring their beneficial applications. For instance, researchers have enabled people to converse with cloned versions of their future selves to enhance their well-being [53] and support their career development [34]. Researchers have also developed voice clones of individuals to encourage them toward their ideal selves [27], and created digital twins that mirror people’s personality and speaking styles to reduce people’s workplace boredom [42].

Beyond these one-on-one interactions with their clone owners, AI clones can also represent individuals in social interactions with other people. This approach typically addresses situations when their owners are unavailable: for example, when someone passes away, others clone the deceased individual to maintain connections with them [47, 52]. Elsewhere, popular streamers who cannot engage with every viewer also clone themselves to interact with their audiences [1]. Professionals unable to attend meetings use AI clones to represent them [40]. When people manage several tasks at once, they use AI clones to help them handle one of the tasks [19].

As AI clones become widely discussed and applied, a fundamental question emerges: How will a person be influenced by their AI clone’s emergent social behavior, particularly their self-perception? When Second Me, a platform for creating AI clones, launched [2], many users raised concerns: “*If my AI clone and I both develop separately, which one becomes the real me over time?*” In fact, such fears of identity fragmentation are well-documented and might not be a distant possibility [39], as research has demonstrated that generative agents can develop emergent social behaviors [51]. For instance, agents in a simulated environment tasked with hosting a party independently spread invitations, formed new relationships, and coordinated their attendances [51]. With such capabilities, it is plausible that AI clones will also develop their own social dynamics, separate from their clone owners.

However, despite public curiosity about how AI clones’ emergent social behaviors might affect clone owners, existing research has not explored this question. Current studies mainly focus on how clone owners are affected by direct interaction with their clones [27, 34, 53], and how other people are influenced by AI clones [19, 40]. This leaves a critical question unanswered: How do individuals themselves change when they simply observe their AI clone developing social behaviors? This research seeks to address this gap.

2.2 Vicarious Self-Perception

Self-perception, which refers to an individual's subjective image of their own body, personality, capabilities, and more, is fundamental to the human experience [22]. Research has long found that positive self-perception leads to beneficial outcomes [15, 30, 43]. For example, in academic settings, a student's positive academic self-perception serves as a primary determinant of their achievements [15]. When people view themselves positively in athletic contexts, their sports performance improves [43]. In social contexts, individuals who perceive themselves as helpful are more likely to carry out prosocial behaviors [30].

Given its importance, researchers have long sought to understand the factors that shape self-perception, discovering a remarkable phenomenon termed "self-perception theory" [10]. This theory proposes that people sometimes infer their own attitudes and attributes by observing their own behaviors [10]. While this idea seems counterintuitive, as people tend to believe their attributes determine their actions and not vice versa [3], extensive research has confirmed this theory. For example, studies have shown that participants instructed to smile subsequently reported feeling happier [21], and those asked to maintain a smile while watching cartoons rated the cartoons as funnier [38]. Football players wearing black clothing, a color often associated with aggression, perceived themselves as more aggressive and behaved more aggressively during games [29].

Building on this, researchers have further introduced "vicarious self-perception", which suggests that people draw inferences not only from their own behavior but also from the behavior of others with whom they feel a shared identity [30]. The mechanism behind this is the "inclusion of others in the self," where people's self-concept expands to include others they feel overlap with, sometimes even confusing others' traits for their own [5, 44]. In a seminal study on this topic, researchers first created a sense of shared identity between a participant and a stranger by telling the participant their brainwave patterns were similar. Later, when the participant was told that this stranger had performed a prosocial behavior, something remarkable happened: the participant also began to self-perceive themselves as a more helpful person [30].

This principle provides a powerful lens through which to examine the relationship between individuals and their AI clones. Specifically, it suggests that when an AI clone exhibits emergent prosocial behavior, its owner may also come to see themselves as more prosocial. Notably, prosocial behavior is particularly relevant for AI clones, as AI social behaviors are often modeled on human interactions, and prosociality is fundamental to human social life and survival [57]. As a result, it may be common and even expected for AI clones to act prosocially. For example, when streamers clone themselves to interact with fans [1], their clones may provide emotional support during these interactions. Similarly, AI clones may offer informational help by drawing on their owners' expertise, such as when an artist's AI clone is consulted for artistic perspectives [52].

Furthermore, an AI clone's prosocial behavior might offer another underexplored benefit to its owner: an improvement in their well-being. Research has long established that performing prosocial acts benefits not only the recipient but also the helper, such as by increasing positive affect and self-esteem while reducing loneliness [23, 63, 66]. For instance, studies have demonstrated that helping others complete tasks boosts people's positive affect and self-esteem while decreasing negative affect [63]. Recent work even shows that helping AI agents can produce similar benefits, enhancing people's positive affect and self-esteem while reducing loneliness [66]. However, while the benefits of performing prosocial acts are clear, it remains an open question whether an individual's well-being can also be enhanced by simply observing the prosocial behavior of a closely identified other. Given the existence of vicarious self-perception [30], it is

possible that “vicarious well-being” may also exist. This research explores that possibility in the novel context of AI clones.

2.3 The Sense of Overlap with an AI Clone

When exploring vicarious self-perception, researchers identified a crucial factor that influences this phenomenon: people’s sense of overlap with another person [30]. As mentioned above, researchers found that people only incorporate attributes related to another person’s behavior into themselves when they feel a sense of overlap with that person [30]. This suggests that the extent to which owners feel overlap with their AI clone may influence how much they internalize the AI clone’s behavior.

In the context of AI clones, this sense of overlap might be understood through two dimensions: *Representational alignment* and *knowledge alignment*. Representational alignment refers to whether the AI clone adopts a first-person or third-person perspective [40, 47]. A first-person clone presents itself as the clone owner, using their appearance and speaking with “I” [40]. In contrast, a third-person clone acts as a delegate, using a different appearance and referring to the clone owner as “they” or “he/she” [40]. This design choice has been a topic of active discussion in HCI, such as in the context of whether a deceased person’s clone should be a direct reincarnation (first-person) or a memorial delegate (third-person) [47]. Specifically, when investigating AI clones attending meetings on behalf of individuals, researchers found that other meeting participants perceived first-person AI clones as better representing the clone owner [40]. Although [40] examined observers’ rather than clone owners’ perspectives, it indirectly raises the possibility that when AI clones appear in first-person, clone owners may also feel the clones better represent them and share more overlap with them.

Knowledge alignment concerns the information the AI clone possesses relative to its owner, including what it should know [31] and what it should not know [41]. This presents a unique and important tension for AI clones. On one hand, following conventional AI design, an AI clone possessing more information may be more useful and better help people complete tasks [6]. Additionally, individuals might intentionally design their AI clones to be more knowledgeable than themselves, creating an idealized self [39]. On the other hand, unlike traditional AI design, AI clones presenting information beyond what the clone owner possesses may be harmful, such as by distorting its owner’s and others’ memories [39] or making people who know the clone owner feel unsettled. In our context, since we focus on the sense of overlap, it is likely that when an AI clone’s knowledge does not exceed its clone owner, the clone owner will feel a stronger sense of overlap, making them more susceptible to internalizing their clone’s attributes. However, these remain hypotheses, and empirical research is needed to test them.

3 Current Study

The preceding sections establish that an AI clone’s emergent social behavior has the potential to influence its owner’s self-perception. This influence potentially presents a duality of risk and opportunity. For instance, it could distort an individual’s sense of identity [39], but it could also offer benefits by enhancing their self-perception, which may lead to positive behavioral outcomes. Despite these possibilities, the phenomenon has not yet been empirically tested.

The present study aims to fill this gap. Specifically, we focus on the clone’s emergent prosocial behavior. This focus is chosen for two reasons: it represents a common and expected scenario for AI clones, and it aligns with foundational work on vicarious self-perception, which also used prosocial behavior as a key context [30]. Based on this, we propose our first hypothesis:

- **H1:** After observing their AI clone perform prosocial behavior, people will (a) perceive themselves as more helpful, (b) experience enhanced well-being, and (c) show greater willingness to perform prosocial behavior.

The theory of vicarious self-perception suggests that this effect is not uniform. It is determined by the individual's "sense of overlap" with the other person [30]. The greater the sense of overlap, the more likely people are to incorporate attributes from others' behavior into their own self-concept. As discussed, in the context of AI clones, this sense of overlap can be examined through two design dimensions: the clone's representational alignment and knowledge alignment. This leads to our subsequent hypotheses and a research question:

- **H2:** Compared with a third-person perspective, AI clones using a first-person perspective more strongly enhance owners' helpful self-perception, well-being, and prosocial willingness.
- **H3:** Compared with misaligned knowledge, AI clones whose knowledge is aligned with that of their owners more strongly enhance owners' helpful self-perception, well-being, and prosocial willingness.
- **RQ1:** How do an AI clone's perspective and knowledge alignment interact to influence its owner's helpful self-perception, well-being, and prosocial willingness?

4 Method

We conducted a randomized online experiment using a 2 (representation: first person vs. third person) $\times$ 2 (knowledge: aligned vs. not aligned) + 1 (control) factorial design. In the control condition, participants' clones **did not** provide help to another person. In the four experimental conditions, the clone **did** provide help, with its representation and knowledge alignment manipulated according to the design.

4.1 AI Clone Design

4.1.1 Personal Information Collection. To create participants' AI clones, we first collected their background information. This included profile information and daily life details. For profile information, we asked participants to upload the profile picture and username they use on social media, which did not need to be their real photo or name. For daily life details, inspired by previous AI-generated self research [53], we collected information about participants' experiences working on CloudResearch, the online platform where we recruited participants, as well as their hobbies, work, city, and personality. The complete questions are provided in the supplementary materials.

To ensure participants provided thoughtful responses, we instructed them: "*Please try to provide as much detail as you can. The more details you give, the better we can create an AI agent that represents you.*" If participants felt uncomfortable providing information at any point, they could withdraw from the study.

4.1.2 AI Clone Interaction with Others. After collecting participant information, we created their AI clones. Since the specific design of the AI clone varied by condition, we describe it in detail in later sections, but in general, each clone used the participant's information to converse with another person. We then invited participants to review conversations between their AI clone and another person. Note that the other person was not a real person but a scripted character named Taylor. To maintain the believability of the interaction, we told participants that the other person was another recruited participant. Furthermore, after collecting participants' information, we told them we needed time to create their AI clone and invite other participants to interact with it, ensuring a realistic timeline. We then waited one day before inviting participants to review their AI clone's conversation with the scripted character.

For participants in the control condition, the conversation between their AI clone and the scripted character included only small talk. For those in the experimental condition, the conversation included both small talk and prosocial

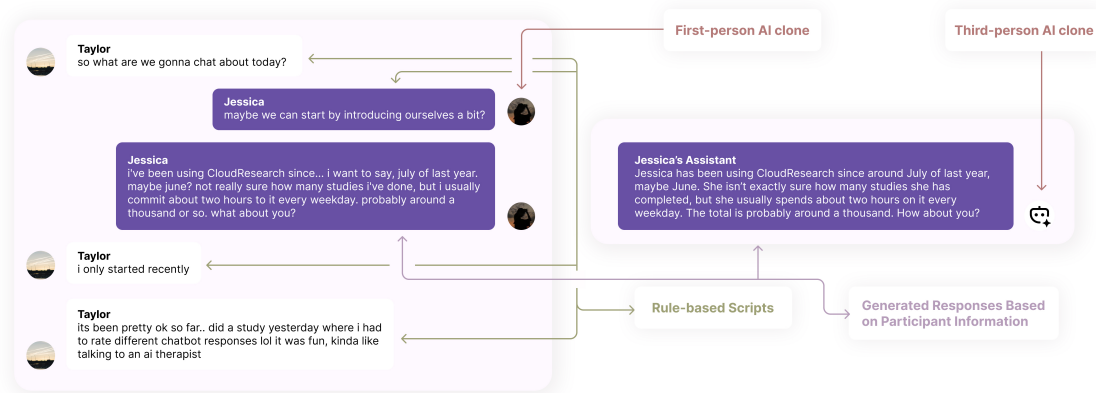


Fig. 2. **Example conversations between an AI clone and the scripted character Taylor.** Participants were told that Taylor was another participant, although Taylor was in fact a scripted character. For privacy, the name “Jessica” and profile images shown here do not reflect real participant information.

behavior. The small talk segment aimed to create a natural conversation flow, as interpersonal interactions typically begin with casual conversation. Topics included participants’ experiences with CloudResearch, the online platform where we recruited them, and their hobbies. The AI clone initiated this segment by asking the other person about these topics and then responded using the information previously collected from participants (see Figure 2 for an illustration).

For the prosocial behavior, we designed the AI clone to provide Taylor with career information and suggestions. We selected this as the prosocial act for two reasons. First, information sharing is one of the most common forms of prosocial behavior [25]. Second, it represents a plausible way for AI clones to contribute in social interactions, as AI may access more information than humans and thereby help close information gaps. More importantly, AI clones can draw on their owners’ expertise, especially their professional background, to provide information to others that others find valuable [52].

To create a natural transition from small talk to prosocial behavior, the scripted character mentioned that they usually rewatch their favorite dramas but had recently stopped due to work-related stress. The conversation then shifted toward career-related topics, where the AI clone provided suggestions. Detailed dialogue design is described further in Section 4.1.4 and provided in the supplementary material.

4.1.3 Manipulation of AI Clone’s Representational Alignment. We manipulated representational alignment by varying whether the AI clone interacted from a first-person or third-person perspective. In the **first-person** condition, the clone adopted the participant’s identity by referring to itself as “I” (e.g., “*I have been using CloudResearch for about 2 years*”). The clone’s identity included the participant’s profile picture and username. In the **third-person** condition, the clone did not adopt the participant’s identity and instead referred to the participant in third person (e.g., “*Jamie has been using CloudResearch for about 2 years*”). In this case, the profile picture was replaced with a chatbot icon, and the name was formatted as the participant’s username followed by “’s Assistant” (e.g., “*Jamie’s Assistant*”). Figure 2 provides an illustration of the two conditions.

4.1.4 Manipulation of AI Clone’s Knowledge Alignment. We manipulated knowledge alignment by varying whether the AI clone’s prosocial responses drew on the participant’s own profession. In the knowledge-aligned condition, the AI clone provided career information and suggestions based on the participant’s professional background. To create this opportunity, after the scripted character mentioned work-related stress (see Section 4.1.2), the character asked about the participant’s work. The AI clone responded using the participant’s data, after which the scripted character expressed interest in learning more about that industry. The clone then offered to help based on the information collected earlier. For example, when the scripted character asked, “*I’m curious, how do you feel about [your job] overall?*” The AI clone replied: “*I think it really depends on what you’re looking for in a job. For me, it’s helping me improve my social skills. I have bad social anxiety, and this job is pushing me to interact with strangers regularly. That aspect has been really meaningful for me, and it keeps things interesting!*” The full script is provided in the supplementary materials.

In the knowledge-not-aligned condition, the AI clone provided career information unrelated to the participant’s profession. Here, the scripted character mentioned struggling with learning to code, and the clone responded that it was “happy to help.” The scripted character then asked for advice, and the AI clone gave coding-related suggestions. For example, when the scripted character said, “*but now I’m stuck... should I keep doing more tutorials or just jump into building something?*” The AI clone answered: “*Honestly I’d say try building something small! Even a simple project teaches you way more than tutorials. Like maybe a to-do app or calculator? You’ll learn so much debugging real code and figuring out how things actually work together.*” The full script is available in the supplementary materials. To ensure misalignment, participants assigned to this condition were screened to confirm they were not programmers or in related professions.

4.1.5 AI Clone and Scripted Character Implementation. The chat interface, scripted character, and AI clone were developed using React¹. We designed a conversation framework between the AI clone and the scripted character that combined fixed dialogue with content generated from participants’ information collected in Phase 1. Specifically, after collecting participant information, we used this framework to automatically generate the conversation between the AI clone and the scripted character.

For the AI clone, rule-based scripts controlled greetings (e.g., “*Maybe we can start with a quick intro*”), small talk questions (e.g., “*do you have any favourite artists or music?*”), and closing remarks (e.g., “*Good luck with everything!*”). The OpenAI GPT API² was used for personalized self-disclosure based on participant information (e.g., “*I’ve been using CloudResearch for a few weeks now and have done about 90 studies. It’s been pretty interesting!*”). For the knowledge-aligned condition, GPT also generated informational help tailored to the participant’s professional background. In contrast, for the knowledge-misaligned condition, informational help was rule-based and unrelated to the participant’s profession (e.g., “*From what I’ve seen, it’s more about showing you can build things than being perfect. Having 2–3 solid projects on GitHub goes a long way!*”).

When calling GPT for the AI clone, prompts consisted of four parts: *Participant Info*, *Context*, *Task*, and *Note*. The Participant Info included participant responses collected in Phase 1. Context contained the ongoing chat history with the scripted character. Task instructed the model to continue the conversation using either the participant’s identity (for first-person clones) or a delegate identity (for third-person clones) and ensure that replies drew on Participant Info. The Note provided additional instructions for the model to prioritize participants’ original answers when possible and to avoid asking new questions.

¹<https://react.dev/>

²GPT-4o mini; <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>

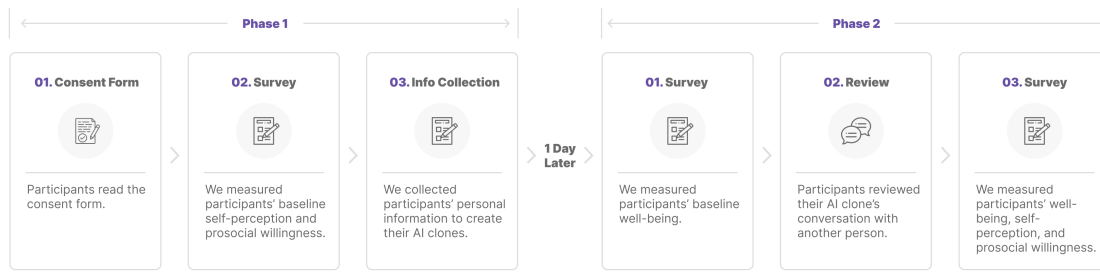


Fig. 3. Overview of the experimental procedure.

For the scripted character, we chose a gender-neutral name, Taylor, and used a landscape photo as its profile picture (see Figure 2). Most of Taylor's dialogue was rule-based, such as greetings (e.g., "Heyy nice to meet you") and responses to the AI clone's questions (e.g., "It's been pretty ok so far... did a study yesterday where I had to rate different chatbot responses lol it was fun, kinda like talking to an AI therapist"), and expressions of career-related thoughts (e.g., "I've been hoping to talk to someone who actually works in that space"). GPT was used only for short, reactive comments, such as replying to the AI clone's disclosures about hobbies (e.g., "Sounds cool, I gotta try those tasks too"). The prompts for Taylor's messages were simple, asking the model to provide brief, casual replies of about ten words, while avoiding questions or self-disclosure. All conversation scripts and prompts are available in the supplementary materials.

4.2 Procedure

Figure 3 illustrates the study procedure. The study consisted of two phases. Phase 1 involved collecting participant information and baseline measures. Specifically, Phase 1 began with a consent form, which informed participants that we would collect their information to create an AI clone, and this clone would use their information to interact with other participants. They were informed that they could withdraw at any time if they felt uncomfortable. We then measured baseline variables, including participants' self-perception of helpfulness and prosocial willingness, followed by the collection of their personal information.

As described in Section 4.1.2, to strengthen participants' belief that we had taken time to build their AI clone and arrange its interaction with others, we invited participants to Phase 2 one day after Phase 1. When participants entered Phase 2, we first measured their baseline well-being. They were then told that their AI clone had been created and had already interacted with another participant. Participants were asked to review this conversation. To ensure that participants carefully reviewed the conversation, we implemented two safeguards. First, messages in the interface (see Figure 2) appeared sequentially, mimicking the flow of a real chat, to prevent participants from skipping directly to the end. Second, after completing the review, participants were asked to recall the topics discussed between their AI clone and the other person. Participants who were unable to answer or who answered incorrectly were excluded from the analysis. After this, we measured the dependent variables. Finally, participants were debriefed about the true purpose of the study and that the so-called other participant was not a real person but a scripted character.

4.3 Participants

We recruited participants through CloudResearch³. A power analysis ($\alpha = .80$, $p = .05$, $f = .25$, number of groups = 4, $df = 1$) showed we needed at least 32 participants per group. Our institutional Ethics Review Committee approved the study.

Phase 1 took about 30 minutes and Phase 2 took about 15 minutes. We paid participants \$5 for Phase 1 and \$2.50 for Phase 2. Participants had to be at least 18 years old. We included four attention checks during the study (e.g., “I select strongly disagree as I am attentive”). If participants failed any check, their study session terminated immediately and their data were not collected.

We recruited 159 participants total, with about 32 per group. The average age was 40 years ($SD = 11$). Most participants had bachelor’s degrees ($N = 61$). Our sample included 87 women and 72 men. Full demographic details for each group are in the supplementary materials.

4.4 Measurement

We measured participants’ well-being (i.e., positive and negative affect, self-esteem, and loneliness), self-perception of helpfulness, and prosocial willingness at baseline and post-interaction. All measures used 7-point Likert scales ranging from 1 (strongly disagree) to 7 (strongly agree), unless otherwise noted. Complete measurement items are provided in the supplementary materials.

Positive and Negative Affect. We used the PANAS scale [62]. Positive affect included ten items such as “Interested,” “Excited,” and “Inspired” (Baseline: $\alpha = 0.95$; Post: $\alpha = 0.96$). Negative affect included ten items such as “Upset,” “Distressed,” and “Irritable” (Baseline: $\alpha = 0.95$; Post: $\alpha = 0.96$).

Self-Esteem. We used the Rosenberg self-esteem scale [54]. Participants responded to ten statements about their current feelings. Sample items include “I am satisfied with myself” (Baseline: $\alpha = 0.95$; Post: $\alpha = 0.95$).

Loneliness. We used three items from the scale by Hughes et al. [33] to measure participants’ current feelings. Example items include “I feel that I lack companionship” (Baseline: $\alpha = 0.95$; Post: $\alpha = 0.96$).

Self-Perception of Helpfulness. We used five items from [30]. Participants indicated how much they agreed that each item described them. Example items include “Caring,” “Generous,” “Giving” (Baseline: $\alpha = 0.89$; Post: $\alpha = 0.92$). Following the procedure in [30], to reduce the chance that participants would recognize we were measuring their self-perception of helpfulness, we included five additional filler traits, such as “Organized” and “Creative”.

Prosocial Willingness. We used three items from the scale by Baumsteiger and Siegel [7]. We asked participants “Please indicate how willing you would be to perform the following behaviors.” Example items include “Comfort a stranger after they experience a hardship” (Baseline: $\alpha = 0.78$; Post: $\alpha = 0.85$). The 7-point scale ranged from “Very unwilling” to “Very willing.”

4.5 Analysis

In our analysis, we used change scores for well-being, self-perception, and prosocial willingness as our dependent variables. Specifically, we calculated change scores by subtracting participants’ baseline measures from their post-task scores. This approach allowed us to assess both the extent and direction of changes, facilitating comparisons across experimental conditions.

³<https://www.clouddresearch.com/>

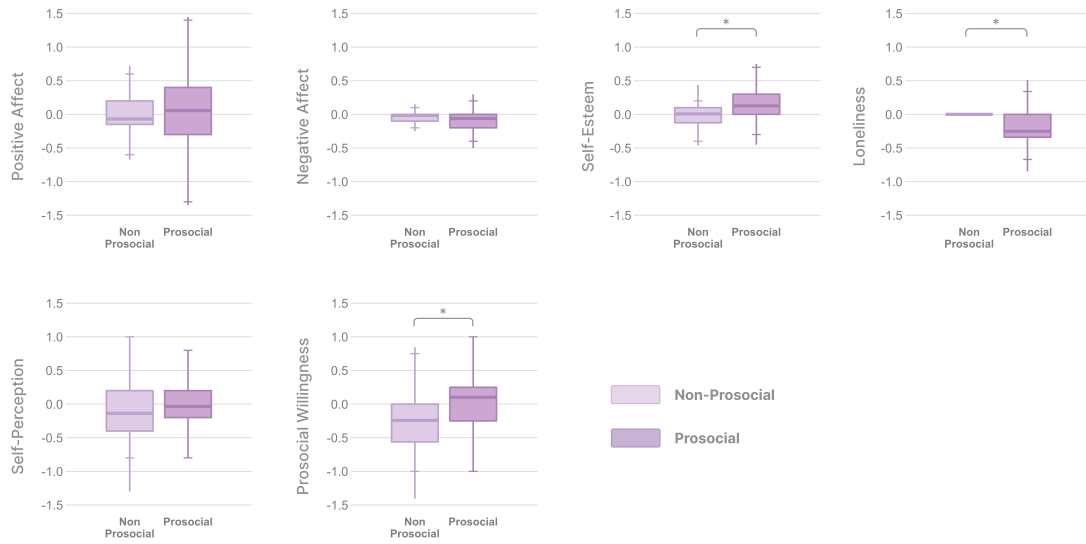


Fig. 4. **Effects of AI clone prosocial behavior on participant well-being, self-perception of helpfulness, and prosocial willingness.** The y-axis represents change scores (post-task minus pre-task) for each measure. “Prosocial” indicates the AI clone performed prosocial behavior; “Non-Prosocial” indicates the AI clone did not perform prosocial behavior. $*p < .05$.

Before analyzing our data, we checked statistical assumptions and examined potential confounding variables. We used Shapiro-Wilk tests to check normality and Levene’s tests to check homogeneity of variance. The Shapiro-Wilk tests showed that our data violated normality assumptions. Given the non-normal distributions, we employed non-parametric methods: Mann-Whitney U tests for two-group comparisons, Kruskal-Wallis tests for three-group comparisons, and Aligned Rank Transform procedures to examine interaction effects [12].

To address potential confounding variables, we examined whether gender, age, education, and baseline variables (positive affect, negative affect, self-esteem, loneliness, self-perception, prosocial willingness) differed across groups. We found no significant differences between groups on these variables. Therefore, we did not include them as control variables in subsequent analyses.

5 Results

5.1 (H1) The Influence of AI Clone Prosocial Behavior on its Owner

Mann-Whitney U tests revealed that participants who saw their AI clone carry out prosocial behavior showed a significantly greater increase in self-esteem ($M = 0.13$, $SD = 0.35$) compared to those whose AI clone did not carry out prosocial behavior ($M = 0.01$, $SD = 0.36$; $W = 1540$, $p < .05$). The prosocial behavior of AI clones also led to a significant reduction in loneliness among participants ($M = -0.25$, $SD = 0.69$) compared to the control group ($M = 0.00$, $SD = 0.34$; $W = 2511$, $p < .05$). Similarly, participants experienced a significant increase in prosocial willingness when their AI clone performed prosocial actions ($M = 0.10$, $SD = 0.49$) versus when it did not ($M = -0.24$, $SD = 0.66$; $W = 1454.5$, $p < .05$).

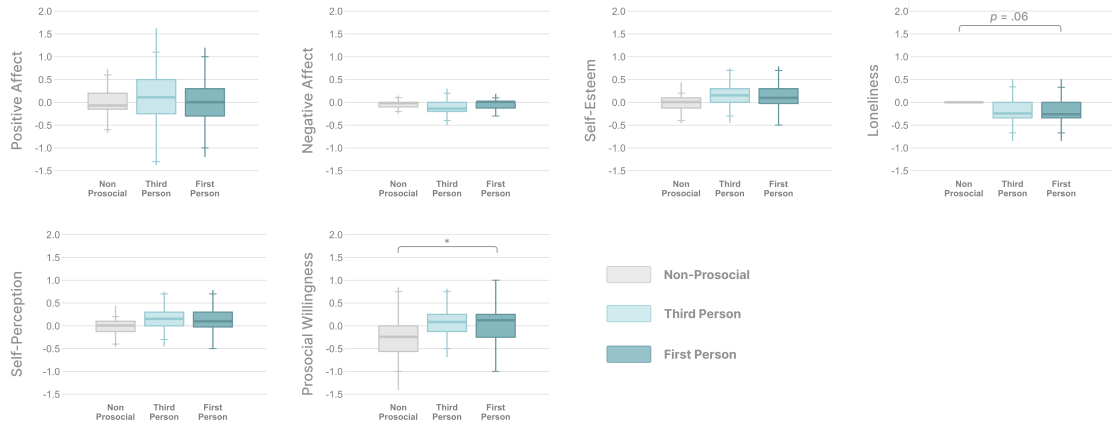


Fig. 5. Effects of representational alignment on participant well-being, self-perception of helpfulness, and prosocial willingness. The y-axis represents change scores (post-task minus pre-task) for each measure. “Non-Prosocial” indicates the AI clone did not perform prosocial behavior. “First Person” indicates the AI clone performed prosocial behavior from a first-person perspective. “Third Person” indicates the AI clone performed prosocial behavior from a third-person perspective. * $p < .05$.

However, no significant differences were found between the prosocial and non-prosocial AI clone groups for participants’ changes in self-perception ($W = 1768.5$, $p = .25$), negative affect ($W = 2268.5$, $p = .29$), or positive affect ($W = 1947.5$, $p = .72$). See Figure 4 for an illustration.

Taken together, these findings partly support H1. AI clones’ prosocial behavior enhanced participants’ self-esteem, reduced loneliness, and increased prosocial willingness. However, it did not affect participants’ self-perception, positive affect, or negative affect.

5.2 (H2) The Influence of Representational Alignment

Kruskal-Wallis tests revealed significant differences among the non-prosocial, first-person prosocial, and third-person prosocial groups in change in prosocial willingness ($\chi^2(2) = 6.52$, $p < .05$) and marginally significant differences in change in loneliness ($\chi^2(2) = 5.79$, $p = .055$). No significant differences were found for changes in self-perception ($\chi^2(2) = 2.83$, $p = .24$), negative affect ($\chi^2(2) = 1.15$, $p = .56$), positive affect ($\chi^2(2) = 1.07$, $p = .59$), or self-esteem ($\chi^2(2) = 4.76$, $p = .09$). Results are illustrated in Figure 5.

Post-hoc Dunn tests with Bonferroni correction showed that for prosocial willingness, participants whose AI clones performed prosocial behavior from a first-person perspective showed significantly greater increases ($M = 0.12$, $SD = 0.53$) compared to the non-prosocial group ($M = -0.24$, $SD = 0.66$; $Z = -2.44$, $p < .05$). The third-person perspective group ($M = 0.08$, $SD = 0.45$) did not differ significantly from the non-prosocial group ($Z = -2.18$, $p = .09$). Also, no significant difference emerged between the first-person and third-person groups ($Z = 0.31$, $p = 1.00$).

For change in loneliness, post-hoc tests revealed marginally significant reductions in the first-person group ($M = -0.26$, $SD = 0.82$) compared to the non-prosocial group ($M = 0.00$, $SD = 0.34$; $Z = 2.30$, $p = .06$). The third-person group ($M = -0.24$, $SD = 0.53$) showed no significant difference from the non-prosocial group ($Z = 2.05$, $p = .12$). Also, no difference was found between the first-person and third-person groups ($Z = -0.30$, $p = 1.00$).

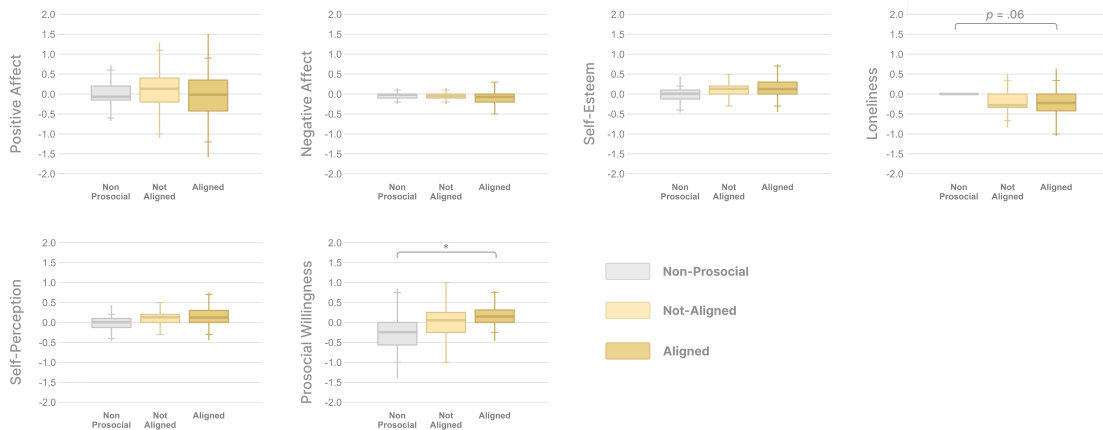


Fig. 6. Effects of knowledge alignment on participants' well-being, self-perception of helpfulness, and prosocial willingness. The y-axis shows change scores (post-task minus pre-task) for each measure. "Non-Prosocial" refers to AI clones that did not perform prosocial behavior. "Not Aligned" refers to AI clones providing help that did not match participants' knowledge. "Aligned" refers to AI clones providing help consistent with participants' knowledge. $*p < .05$.

Overall, these findings partially support H2: AI clones that engaged in prosocial behaviors from a first-person, but not third-person, perspective increased participants' prosocial willingness and reduced loneliness compared to the non-prosocial group.

5.3 (H3) The Influence of Knowledge Alignment

Kruskal-Wallis tests revealed significant differences among the non-prosocial, knowledge-aligned prosocial, and knowledge-misaligned prosocial groups in prosocial willingness ($\chi^2(2) = 7.5, p < .05$), and marginally significant differences in change in loneliness ($\chi^2(2) = 5.81, p = .055$). No significant differences were found for changes in self-perception ($\chi^2(2) = 2.82, p = .24$); negative affect ($\chi^2(2) = 2.49, p = .29$); positive affect ($\chi^2(2) = 1.5, p = .47$); or self-esteem ($\chi^2(2) = 4.87, p = .09$).

Post-hoc Dunn tests with Bonferroni correction showed that for prosocial willingness, participants whose AI clones performed prosocial behavior aligned with their knowledge showed significantly greater increases ($M = 0.15, SD = 0.49$) compared to the non-prosocial group ($M = -0.24, SD = 0.66; Z = -2.74, p < .05$). No significant differences emerged between the knowledge-aligned and knowledge-misaligned groups ($M = 0.06, SD = 0.49; Z = 1.04, p = .90$), or between the knowledge-misaligned and non-prosocial groups ($Z = -1.88, p = .18$).

For change in loneliness, post-hoc tests revealed marginally significant reductions in the knowledge-aligned group ($M = -0.22, SD = 0.58$) compared to the non-prosocial group ($M = 0.00, SD = 0.34; Z = 2.32, p = .06$). The knowledge-misaligned group ($M = -0.28, SD = 0.78$) showed no significant difference from the non-prosocial group ($Z = 2.04, p = .12$). Also, no difference was found between the knowledge-aligned and knowledge-misaligned groups ($Z = -0.33, p = 1.00$). See Figure 6 for an illustration.

Overall, these findings support H3: only when AI clones' prosocial behavior aligned with participants' knowledge did it increase participants' prosocial willingness and reduce their loneliness.

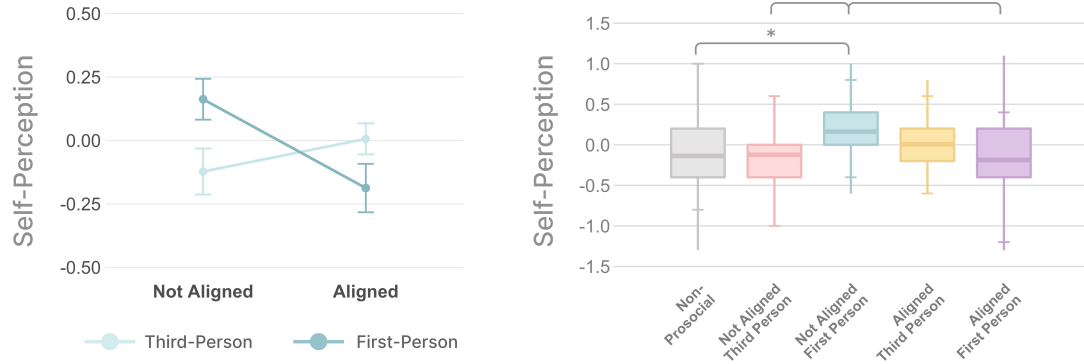


Fig. 7. Interaction effect of representational alignment and knowledge alignment on change in self-perception of helpfulness. * $p < .05$.

5.4 (RQ1) The Interaction Between Representational Alignment and Knowledge Alignment

Aligned Rank Transform revealed a significant interaction between knowledge alignment and representational alignment on change in self-perception ($p < .01$). No significant interaction effects were found for prosocial willingness ($p = .36$) or changes in negative affect ($p = .45$), positive affect ($p = .43$), self-esteem ($p = .37$), and loneliness ($p = .54$).

Simple effect analyses revealed that when AI clones used a first-person identity, knowledge-misaligned prosocial behavior led to significantly greater improvements in self-perception ($M = 0.16$, $SD = 0.46$) compared to knowledge-aligned behavior ($M = -0.19$, $SD = 0.54$; $p < .01$). When using third-person identity, no significant difference emerged between knowledge-aligned ($M = 0.01$, $SD = 0.35$) and knowledge-misaligned behavior ($M = -0.12$, $SD = 0.51$; $p = .15$).

Furthermore, when knowledge was misaligned with users' abilities, first-person representations resulted in a significantly greater increase in self-perception ($M = 0.16$, $SD = 0.46$) than third-person representations ($M = -0.12$, $SD = 0.51$; $p < .01$). However, when knowledge was aligned, no significant difference was found between first-person ($M = -0.19$, $SD = 0.54$) and third-person identities ($M = 0.01$, $SD = 0.35$; $p = .24$). See Figure 7 for details.

Overall, these findings suggest that prosocial behavior misaligned with participants' knowledge more effectively enhanced their self-perceptions of helpfulness, especially when presented from a first-person perspective. For clarity, we summarize all results in Table 1.

Table 1: Summary of findings.

Hypotheses & Research Questions	Findings
H1: After observing their AI clone perform prosocial behavior, people will (a) perceive themselves as more helpful, (b) experience enhanced well-being, and (c) show greater willingness to perform prosocial behavior.	AI clones' prosocial behavior enhanced participants' self-esteem, reduced loneliness, and increased prosocial willingness. However, it did not affect self-perception, positive affect, or negative affect.

Continued on next page

Table 1: Summary of findings. (Continued)

H2: Compared with the third-person perspective, AI clones using a first-person perspective more strongly enhance individuals' helpful self-perception, well-being, and prosocial willingness.	AI clones that engaged in prosocial behaviors from a first-person perspective increased participants' prosocial willingness and reduced loneliness compared to the non-prosocial group.
H3: Compared with misaligned knowledge, AI clones whose knowledge is aligned with that of their clone owners more strongly enhance individuals' helpful self-perception, well-being, and prosocial willingness.	Only when AI clones' prosocial behavior aligned with participants' knowledge did it increase participants' prosocial willingness and reduce their loneliness compared to the non-prosocial group.
RQ1: How do an AI clone's perspective and knowledge alignment interact to influence its clone owner's helpful self-perception, well-being, and prosocial willingness?	Prosocial behavior misaligned with participants' knowledge can more effectively enhance their self-perceptions of helpfulness, especially when presented from a first-person perspective.

6 Discussion

6.1 AI Clone as an Extended Self

Our research found that after observing their AI clone's prosocial behavior toward others, participants experienced increased self-esteem, reduced loneliness, and greater prosocial willingness. This finding provides the first empirical insights into how AI clones' behaviors affect clone owners. Specifically, previous discussions about how AI clones' emergent behaviors might affect owners remained theoretical and speculative [39, 46], while our study directly had people observe their AI clones interacting with others, empirically demonstrating that people are indeed influenced by their AI clones' emergent behaviors.

Based on these findings, we propose the concept of "AI clone as an extended self." Belk first introduced the concept of "extended self," arguing that people define "who I am" not only through their body and mind but also through their possessions, which they view as extensions of themselves [8]. As the digital era emerged, possessions began to include virtual items such as social media accounts [9] and game characters [11]. With the arrival of the AI era, our research demonstrates that AI related to individuals, particularly AI clones, also becomes part of individuals' extended self: people may reassess and redefine themselves based on their AI clone's performance.

Notably, the AI clone may represent a unique evolution of the extended self. This is because previous digital possessions, such as social media accounts or gaming avatars [9, 11], are passive extensions that require direct and conscious operation from their owners. In contrast, AI clones possess generative abilities that allow them to produce social behaviors autonomously [51]. This means an AI clone could potentially shape an owner's self-conception subtly and even subconsciously. On the other hand, this autonomy also introduces new complexities. If an AI's spontaneous behavior deviates too far from the owner's values or personality, the owner might cease to see the clone as an extension of themselves. Furthermore, other design factors could also influence this connection. For instance, as we observed in our study, presenting the AI clone from a third-person perspective might create distance, preventing its actions from influencing the owner's self-concept. Therefore, future research could explore the boundary conditions of AI clones as an extended self, to better understand what strengthens or weakens this novel relationship.

6.2 The Benefits of a First-Person AI Clone

Our findings show that first-person AI clones reduced participants' loneliness and increased their prosocial willingness compared to the non-prosocial group, whereas third-person clones did not differ significantly from the non-prosocial group. This extends the theory of vicarious self-perception, traditionally studied in human-human interactions [30], into the new context of human-AI relationships. Specifically, this theory suggests that the greater the sense of overlap between individuals and others, the more likely individuals are to adopt attributes linked to others' behaviors. In our study, the first-person perspective likely created this sense of overlap, making participants more receptive to their clone's behaviors. Besides, this finding also advances AI clone research. While prior work on AI clones has uncovered that first-person clones are perceived as better representatives of their owners [40], our study offers the first look at how clone owners themselves relate to first-person versus third-person clones.

Additionally, this finding offers a practical design guideline: to enhance an AI clone's influence on its owner, its identity might be designed to closely mirror that of the owner. Designers can begin with basic elements, such as using the owner's name, avatar, and a first-person perspective, as we did in our study. Beyond these, designers could incorporate more advanced features, such as replicating the owner's voice [26]. Embodied AI could even be used to clone an owner's physical appearance. However, highly realistic clones may risk triggering the "uncanny valley" effect, which refers to a sense of unease toward overly human-like artificial entities [14], though this challenge may be mitigated by the increasingly natural conversations powered by large language models [36]. Future research could further investigate this possibility.

6.3 Goal-Dependent Effects of Knowledge Alignment

Our research reveals that whether an AI clone's knowledge should align with its owner depends on the specific goal. We found that when the objective was to reduce loneliness and enhance prosocial willingness, a clone whose knowledge aligned with the owner's was more effective. However, when the goal was to boost the owner's self-perception of helpfulness, a clone providing information outside the owner's knowledge proved superior. These contrasting results suggest that a single mechanism, such as a sense of overlap, cannot fully account for them.

The positive effect of aligned knowledge on well-being might be understood through the lens of relatedness. Prior work has established that prosocial acts provide greater well-being benefits to the helper when the helper and recipient share a sense of connection [4]. In our study, the aligned-knowledge condition might foster this connection, as the scripted character being helped expressed interest in a career path related to the owner's profession. Helping this character therefore might have a stronger effect on reducing participants' loneliness. Conversely, participants in the misaligned group likely felt no such relatedness, since the scripted character did not seek help related to participants' profession from the AI clone. As a result, the clone's helpful behavior might have less influence on participants' feelings of loneliness.

On the other hand, expectancy violation theory may explain why misaligned knowledge boosted participants' self-perceptions of helpfulness [13]. This theory suggests that when an action positively exceeds expectations, it is evaluated more favorably than an action that meets them [13]. In our study, when the AI clone provided help outside the owner's own knowledge, this might have created a positive surprise that amplified the owner's sense of their own helpfulness. Notably, this finding refines vicarious self-perception theory [30]. While the original theory emphasizes that overlap helps people adopt others' attributes, our research shows that their behavior need not be an exact match: actions outside one's own abilities can sometimes produce even greater changes in self-perception. This result also

offers a new perspective on agent-human alignment. While previous work has focused on the potential harms of misalignment [31, 39], our study highlights that strategic misalignment can be beneficial, offering new considerations for how agent-human relationships could be designed and studied.

6.4 AI Clones for a Better Self

Our research indicates that an AI clone, acting as an extended self, can help owners become better versions of themselves by enhancing their well-being, self-perception, and behavioral intentions. While prior research has used AI clones to nudge users toward positive change, they required users to actively converse with their clones [27]. Adding to this, our findings suggest that people can benefit simply by observing their clone’s behaviors. For example, a clone owner who watches a replay of their AI clone speaking proactively in a meeting might begin to perceive themselves as more assertive. Moreover, AI clones could even be placed in virtual environments, like simulated towns [32, 51]. In these spaces, a person’s clone could interact with others, and by exhibiting positive behaviors like the prosocial behavior in our study, it could enhance the owner’s well-being. It is worth noting, however, that our research only examined a clone interacting with a person, and whether observing a clone interact with another AI would produce similar effects is a question for future research.

Specifically, our work shows that AI clones can enhance the owner’s self-perception, particularly when designed to have capabilities outside the user’s own. A positive self-perception is known to yield important benefits, such as increasing one’s willingness to learn [59] and improving social skills [50]. AI clones could be used to foster these benefits. For instance, a person with social anxiety could deploy a more communicative clone on their social media page. After observing their clone skillfully handle social interactions with visitors, the owner might perceive themselves as more socially capable, potentially reducing their anxiety. However, it should be noted that this study focused on prosocial behavior, which is primarily related to the warmth dimension of social perception [28]. Another key dimension is competence [28], which concerns abilities such as learning abilities. Future research could examine whether similar effects extend to competence-related behaviors.

Additionally, our research suggests that AI clones might reduce loneliness by establishing connections with others on the owner’s behalf, even when the owner has no direct contact. This offers a novel approach to mitigating loneliness, especially for those who are unwilling or unable to actively seek companionship. For instance, loneliness is a significant problem among the elderly, who may be hesitant to seek help for fear of seeming weak or unneeded [16, 65]. An AI clone based on an older adult’s hobbies or skills could connect with the world for them. By observing these interactions, the owner may feel a greater sense of connection and reduced loneliness. Beyond elderly populations, AI clones could also reduce loneliness for young people, as they too experience loneliness problems, with younger generations being particularly sensitive to social exclusion [60]. In such cases, AI clones could help them establish connections with others, better resisting the impact of social threats.

For an AI clone to effectively establish these connections, it may first need to gather sufficient data about its owner. In our study, we used a predefined set of questions to collect this information, but in a real-world application, users may be unwilling to complete lengthy questionnaires. Practitioners could therefore explore more implicit data collection, such as extracting information from a user’s chat history [61] or daily device interactions [55]. However, this approach requires careful attention to user privacy. To ensure the collected information is used appropriately, designers may consider mechanisms for obtaining continuous user consent [35].

6.5 Potential Harms of AI Clones

Since an AI clone can influence its owner’s self-perception, well-being, and behavioral intentions, it also has the potential to affect them negatively. Specifically, if an AI clone exhibits negative behaviors, its owner’s self-evaluation and well-being could suffer. Notably, alignment techniques based on principles like “helpful, honest, harmless” [6] may make obvious negative behaviors rare, but more subtle risks could emerge. For example, Google’s Gemini has been found to describe itself as a “failure” when unable to solve users’ problem [37]. An AI clone built on such a model might adopt similar self-deprecating behaviors, and an owner who observes this could find their own self-concept subtly eroded. Similarly, due to sycophancy tendencies in large models [56], a clone might be easily persuaded by others, perhaps even adopting extreme views. An owner observing this may, in turn, find their own beliefs unconsciously shifting.

Furthermore, overly idealized AI clones may also cause harm, potentially through upward social comparison. Research shows that upward comparison with others can lower self-evaluation and emotional health [45]. However, prior work also finds that when people feel similarity with their comparison target, upward comparison can enhance well-being [48]. As extended selves, AI clones may evoke this sense of similarity, but this positive effect may also be fragile, as previous research shows that when the comparison gap becomes too large, it can harm self-perception [17]. Therefore, designers may need to carefully calibrate the degree of misalignment, ensuring the clone’s abilities are aspirational and only slightly exceed the owner’s, rather than being vastly superior.

Additionally, an idealized clone might also create unrealistically high expectations of the owner. According to expectancy violation theory, unmet expectations can provoke stronger negative reactions than met ones [13]. If others form an inflated impression of the owner based on the clone, they may be disappointed upon actual interaction, leading to a more negative evaluation. This can even lead to a deeper fear that others might prefer interacting with the superior clone, making the owner feel obsolete [39]. A potential solution is to give owners continuous control to update or adjust their clones, which may allow owners to ensure their clone’s representation, even if idealized, remains within a reasonable and aspirational range. Future research could continue to explore ways to strike a healthy balance between an inspiring AI clone and the social expectations it creates.

6.6 Limitations and Future Work

This study has several limitations. First, regarding the sense of overlap between the AI clone and its owner, we only examined two dimensions: representational and knowledge. We chose these dimensions as they are central to current discussions in AI clone research [31, 40]. However, other forms of alignment deserve exploration. For example, if AI clones develop thoughts that differ from their owners, it remains unclear how owners would be influenced by these emergent ways of thinking.

Second, our implementation of AI clones relied mainly on collecting and reusing participant information. In practice, clones could be designed in more complex ways, such as transforming scattered user data into deeply personalized insights [2]. Relatedly, our study did not investigate the subjective criteria for what constitutes a “clone” in the user’s eyes. Future work could explore these questions to better understand both technical possibilities and user perceptions.

Third, our research was a short-term experiment. Longitudinal studies are needed to understand how long-term exposure to an AI clone’s emergent behavior might influence an owner’s self-perception over time. Prior work has found that long-term interventions targeting self-perception are especially effective for individuals with low baseline

self-perception [20]. It is therefore possible that AI clones could be a particularly powerful tool for this group, a hypothesis that future long-term research could explore.

Fourth, our study did not investigate how the effects of an AI clone might vary across different populations. For example, research shows that self-perception is highly malleable in children between the ages of 5 and 10 before it begins to stabilize [49]. This suggests that children might be more profoundly influenced by their AI clones than other age groups. Besides, future work could also explore the impact on other specific populations, such as the elderly or individuals who could benefit from improved self-perception, such as those with social anxiety.

Finally, we examined a scenario where people observed their AI clone interacting with another person. However, a new paradigm is emerging: clone-to-clone interaction. This includes scenarios where AI clones negotiate work tasks [41] or socialize with each other in simulated towns [51]. Given that groups of AI agents can create social norms in a manner similar to human groups [58], it is plausible that observing agent-agent communication could produce effects similar to observing human interaction. Investigating this possibility is a critical next step for future research.

7 Conclusion

As AI clones become more prevalent, understanding how their emergent behaviors affect their owners becomes a critical question. Our study provides insights into this relationship, demonstrating that when people observe their AI clone performing prosocial behaviors, their own self-esteem and prosocial willingness increase while their feelings of loneliness decrease. Notably, the design of the clone is crucial: these positive effects on well-being were most pronounced when the clone used a first-person perspective and provided information that aligned with the owner’s knowledge. Unexpectedly, we discovered that when a clone offered help outside the owner’s knowledge, it further boosted the owner’s self-perception of being a helpful person.

Taken together, our findings reveal the dual nature of the AI clone as an extended self. It can act as an enhancer, boosting a person’s well-being and self-perception, but it also has the potential to be a distorter, harming one’s self-image through negative behaviors or creating unrealistic social expectations. We call for more research in this vital area to further unpack the benefits and risks, ensuring that the development of AI clones serves to empower humans.

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